

Math364

Principal Component Analysis and Six-Month Forecasting of Global Government Yield Curves

U.S., Euro-Area, and Japanese Bond Markets

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Data sample through April 24, 2026

Abstract

This project studies whether a small number of statistical factors can explain and forecast movements in major government yield curves. We collect daily U.S. Treasury, euro-area government-bond, and Japanese government-bond yield-curve data, align the markets to a common set of maturities, and run principal component analysis (PCA) on weekly yield changes. We then use the first five global PCA scores in a linear regression model to forecast the yield curves six months ahead.

The main result is interpretability rather than strong predictability. Region-by-region, the first component is a level factor, the second is a slope factor, and the third is generally a curvature factor. These findings match the classic empirical structure of fixed-income risk: most curve movement is parallel repricing, while smaller components describe steepening/flattening and belly-versus-wing moves. The global PCA also reveals regional spread components, showing that U.S., euro-area, and Japanese curves do not always move together.

The six-month forecast is deliberately conservative. Because lagged PCA scores have weak power for predicting future weekly rate changes, the linear regression model produces small yield changes: slightly steeper U.S. and euro-area curves, and modest upward changes in Japan. The low one-week R^2 values are not a coding error; they reflect the fact that weekly interest-rate changes are mostly driven by new information rather than by last week's curve movement.

1 Project Goal and Workflow

The goal of the project is to take a large panel of yield-curve observations and reduce it to a small number of economically interpretable factors. This helps answer three questions:

1. Which yield-curve movements explain most weekly rate variation?
2. Are the components similar across the United States, the euro area, and Japan?
3. Can the first five PCA factors be used in a simple linear model to predict the curves six months forward?

The workflow is fully reproducible. First, the script downloads the source data and converts each market to a daily wide table. Second, it keeps a common tenor set of 2Y, 5Y, 10Y, 20Y, and 30Y. Third, it resamples each series to Friday weekly observations and computes weekly changes in basis points. Fourth, it standardizes every feature and estimates both regional and global PCA models. Finally, it forecasts the first five global PCA scores with ordinary least squares and maps those predicted factor scores back into yield-curve changes.

Sample. The final aligned PCA sample contains 1,053 weekly changes from February 17, 2006 through April 24, 2026. The forecast starts from the April 24, 2026 weekly curve and extends 26 weeks to October 23, 2026.

Table 1: Data sources and modeling choices

Item	Choice	Reason
Markets	United States, euro area, Japan	Captures three major developed bond markets.
Tenors	2Y, 5Y, 10Y, 20Y, 30Y	Common across all three regions and enough to identify level, slope, and curvature.
Frequency	Weekly Friday observations	Reduces daily noise and aligns markets with different holidays.
PCA input	Weekly yield changes in basis points	Avoids persistence in yield levels and focuses on curve moves.
Scaling	Standardize each region-tenor series	Prevents high-volatility tenors or markets from mechanically dominating PCA.
Forecast model	Linear regression on lagged global PC scores	Transparent baseline model that is easy to interpret.

2 Data Preparation and PCA Method

Let $y_{i,t}$ be the yield for feature i at week t , where a feature is one region-tenor pair such as U.S. 10Y or Japan 30Y. The model does not run PCA on $y_{i,t}$ directly. Instead it uses weekly changes:

$$\Delta y_{i,t} = 100(y_{i,t} - y_{i,t-1}),$$

where the factor 100 converts percentage-point changes into basis points. Each change series is then standardized:

$$z_{i,t} = \frac{\Delta y_{i,t} - \mu_i}{\sigma_i}.$$

The standardized data matrix Z has 15 columns: five tenors for each of three regions. PCA finds orthogonal directions $v_1, \dots, v_k$ that maximize variance in the projected scores:

$$s_{j,t} = Z_t v_j.$$

The entries of v_j are called loadings. A loading is the weight assigned to a specific region-tenor series inside one principal component. The sign of an individual component is arbitrary, so the pipeline orients components so level or long-end exposure is positive when possible.

This modeling choice has an important implication. PCA is excellent at describing the structure of rate movements, but PCA alone is not a forecasting model. It tells us which combinations of yield changes are historically important; it does not guarantee that those combinations are predictable. That is why the forecast stage is separated from the decomposition stage.

Table 2: Interpretation guide for PCA loadings

Pattern	Loading shape	Economic interpretation
Level	Same sign across most tenors	Broad rise or fall in yields across the curve.
Slope	Short maturities load against long maturities	Steepening or flattening of the yield curve.
Curvature	Belly maturities load against wings	Butterfly move, often centered around 5Y or 10Y.
Regional spread	One region loads against another	Cross-market divergence between U.S., euro-area, and Japanese rates.
Localized residual	Loadings concentrated in a small set of tenors	Smaller relative-value moves not summarized by level, slope, or curvature.

3 Explained Variance Results

The explained-variance table shows how much standardized weekly-change variation each component captures. For each regional curve, PC1 and PC2 explain nearly all of the variation. This is a common result in fixed-income markets: one factor captures broad rate repricing, while the second captures slope changes.

The U.S. curve has the strongest first component: PC1 explains 85.5% of variance, meaning U.S. weekly moves are especially dominated by parallel repricing. Japan has the smallest PC1 share and the largest PC2 share. This means Japanese curve moves in the sample have relatively more slope behavior. The global PCA is more distributed: PC1 explains 50.6%, and the first five components explain 91.9%. This is expected because the global panel must describe both curve-shape effects and cross-region divergence.

Table 3: Explained variance by PCA component

Scope	PC1	PC2	PC3	PC4	PC5
United States	85.5%	12.5%	1.5%	0.3%	0.2%
Euro area	77.0%	16.0%	5.3%	1.6%	0.2%
Japan	72.0%	20.9%	4.7%	1.4%	0.9%
Global	50.6%	16.1%	11.7%	8.0%	5.5%

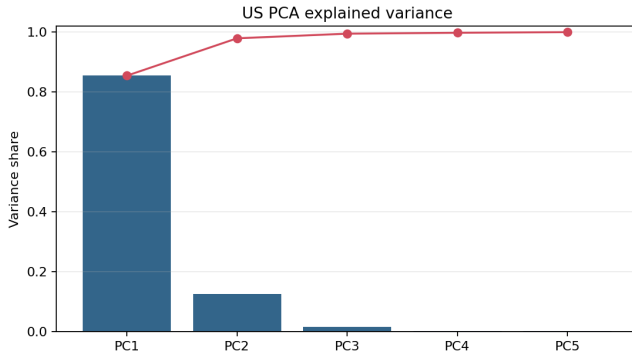


Figure 1: U.S. explained variance. PC1 dominates, with PC2 capturing most of the remaining shape risk.

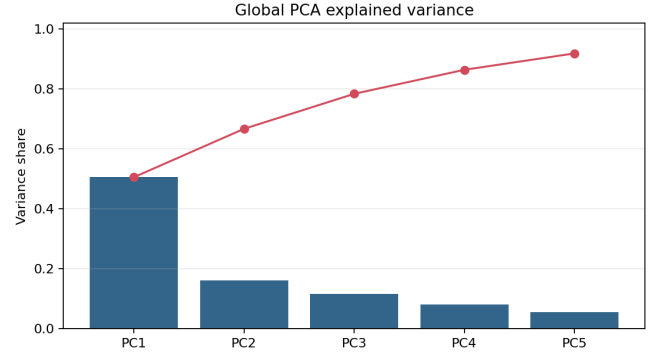


Figure 2: Global explained variance. The global model needs more components because it captures regional divergence as well as curve shape.

4 Regional Loading Interpretation

The loading curves reveal why PC1, PC2, and PC3 can be interpreted as level, slope, and curvature factors. In the U.S. model, PC1 has positive loadings across all maturities and is fairly flat. A positive PC1 score therefore represents a broad upward rate move. PC2 has negative loadings at the short end and positive loadings at the long end, so it represents a steepening move when the score is positive. PC3 emphasizes the difference between the belly and the wings of the curve.

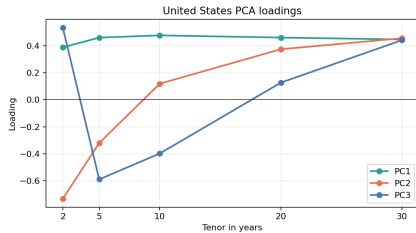


Figure 3: U.S. regional PCA loadings.

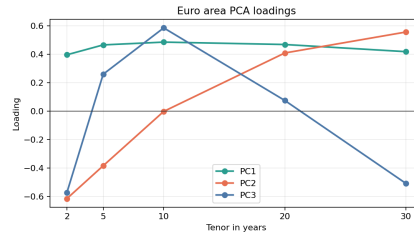


Figure 4: Euro-area regional PCA loadings.

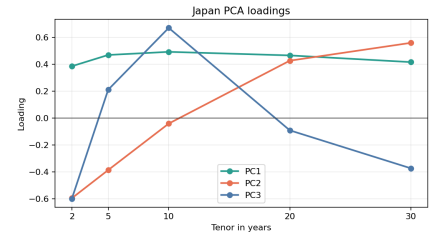


Figure 5: Japan regional PCA loadings.

The euro-area and Japanese loading plots show the same broad hierarchy but with different intensity. Euro-area PC3 is visibly a belly-versus-wing factor, with the 10Y area moving against the short and long ends. Japan has a larger slope component, consistent with the explained-variance table. Across all three regions, the conclusion is stable: fixed-income curve risk is low-dimensional, and most movement can be summarized with three shapes.

What the loading magnitude means. A larger absolute loading means that tenor contributes more to the component score. For example, if the 30Y loading is large and positive in PC2 while the 2Y loading is negative, then a positive PC2 score corresponds to a relative rise in long yields versus short yields. The loading is not itself a yield change; it is a coefficient used to combine standardized yield changes into a PCA factor.

5 Global Components and Cross-Market Risk

The global PCA combines all 15 series into one model. The first global component is still a level factor, but later components are not simply global slope and curvature. Instead, PC2 and PC3 separate regions. This is important because global rates risk is not only about one worldwide curve move; it also includes relative repricing between U.S., euro-area, and Japanese markets.

The heatmap shows that global PC1 has broadly same-direction exposure, especially to U.S. intermediate and long yields. Global PC2 loads positively on Japan and negatively on the euro area, making it a Japan-versus-Europe spread factor. Global PC3 loads strongly on the U.S. against the euro area, making it a U.S.-versus-Europe spread factor. Global PC4 has a slope-like profile across several markets. These components are useful because a portfolio exposed to “rates” may still behave differently depending on whether global, regional, or curve-shape factors are moving.

6 Forecasting With PCA Factors

After estimating the first five global PCA factors, the forecast model uses a linear regression:

$$\mathbf{s}_{t+1} = \alpha + B\mathbf{s}_t + \varepsilon_{t+1},$$

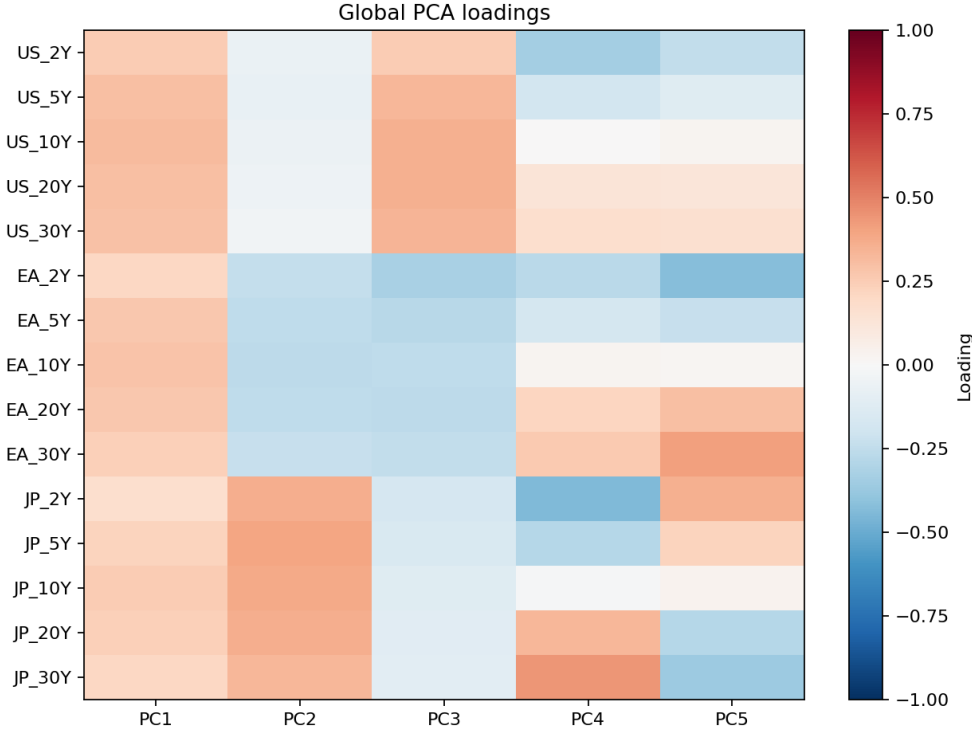


Figure 6: Global PCA loading heatmap. Warm and cool colors identify which region-tenor series move together or against each other within each component.

where $\mathbf{s}_t$ is the vector of the first five global PCA scores. The fitted model forecasts one week ahead. To obtain a six-month forecast, the model is applied recursively for 26 weeks. Each predicted score vector is mapped back through the PCA loadings and standardization parameters to recover predicted weekly yield changes. These weekly changes are then accumulated from the latest observed yield curve.

This is intentionally a simple benchmark. It tests whether last week's factor scores contain useful linear information about next week's factor scores. If the coefficients are close to zero, the forecast naturally drifts toward the historical average weekly change. That is what happens here.

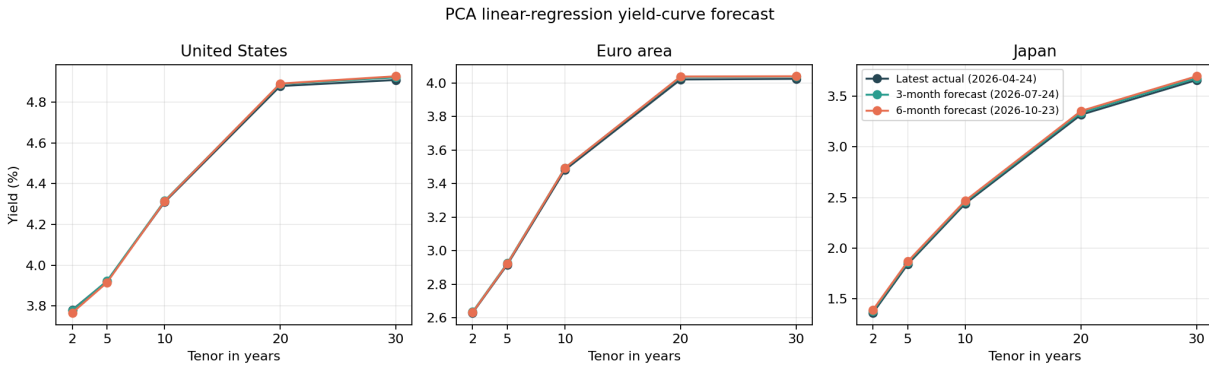


Figure 7: Latest actual curve, three-month forecast, and six-month forecast. The forecast changes are small because the linear factor model finds little persistence in weekly PCA score changes.

7 Forecast Results and Interpretation

The six-month forecast endpoint is shown below. The U.S. forecast implies a very small steepening: the 2Y and 5Y fall slightly while the 20Y and 30Y rise slightly. The euro-area forecast also shows mild upward pressure, especially beyond 10Y. Japan has the largest positive forecast changes, though even there the move is only about 3 to 4 basis points over six months.

Table 4: Six-month forecast endpoint, ending October 23, 2026

Region	Tenor	Latest yield	Forecast yield	Forecast change
U.S.	2Y	3.780%	3.766%	-1.4 bp
U.S.	5Y	3.920%	3.914%	-0.6 bp
U.S.	10Y	4.310%	4.314%	+0.4 bp
U.S.	20Y	4.880%	4.892%	+1.2 bp
U.S.	30Y	4.910%	4.928%	+1.8 bp
Euro area	2Y	2.630%	2.632%	+0.2 bp
Euro area	5Y	2.917%	2.922%	+0.5 bp
Euro area	10Y	3.483%	3.494%	+1.2 bp
Euro area	20Y	4.021%	4.039%	+1.7 bp
Euro area	30Y	4.025%	4.040%	+1.5 bp
Japan	2Y	1.363%	1.393%	+3.0 bp
Japan	5Y	1.842%	1.870%	+2.8 bp
Japan	10Y	2.443%	2.471%	+2.8 bp
Japan	20Y	3.318%	3.355%	+3.7 bp
Japan	30Y	3.660%	3.697%	+3.7 bp

8 Forecast Reliability

The forecast should not be interpreted as a high-conviction prediction of where rates must go. It is better interpreted as a transparent baseline. The in-sample one-week R^2 values are very small:

Table 5: One-week linear regression fit for global PCA scores

Component	One-week R^2	One-week RMSE
PC1	0.010	2.741
PC2	0.010	1.548
PC3	0.022	1.310
PC4	0.016	1.085
PC5	0.008	0.899

The low R^2 occurs because the model is predicting weekly changes, not yield levels. Yield levels are persistent, so a naive model that predicts next week's level from this week's level would look much more accurate. But the economically useful question is whether we can predict the change. Weekly changes in interest rates are heavily influenced by new information: central-bank speeches, inflation releases, labor-market data, auction results, volatility, and risk sentiment. These surprises are not contained very strongly in last week's PCA scores.

This result is not a failure of PCA. PCA is a decomposition method, not a forecasting method. It explains how yields move together after the fact. Forecasting requires information about why future expectations

might change. The linear regression model intentionally uses only lagged PCA scores, so its muted forecast is evidence that lagged curve movement alone has limited predictive power.

9 Conclusion and Extensions

The project produces two main findings. First, the structure of yield-curve risk is highly interpretable. Level risk dominates in all three regions, slope risk is second, and curvature risk is smaller but still meaningful. This matches the standard fixed-income intuition that a small number of factors can describe most government-bond curve movement.

Second, interpretability is not the same as forecastability. The first five global PCA factors explain 91.9% of historical standardized weekly variation, but their lagged values explain very little of next week's factor movement. The six-month forecast is therefore small and mean-reverting. This is a useful baseline because it shows what the yield-curve model predicts when it is given no macro information beyond recent curve movements.

Several extensions would make the forecast more realistic. A stronger model could add policy-rate expectations, futures-implied short rates, inflation surprises, unemployment data, central-bank meeting dates, term-premium estimates, volatility indices, exchange rates, and equity risk indicators. A vector autoregression, dynamic factor model, or state-space model could also be used instead of simple ordinary least squares. For this Math364 project, however, the simple PCA-plus-linear-regression framework has the advantage of being transparent, reproducible, and easy to interpret.

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